Ajinkya K. Ranade, D. Eng.

Senior Technology Scientist | Sustainable Water Treatment

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Thane, India.

SUMMARY

A scientist and former **JSPS Postdoctoral Fellow** with a strong record of high-impact, interdisciplinary research in optoelectronics and **sustainable water treatment for emerging contaminant management**. Deep expertise at the intersection of chemistry, nanotechnology, interface and 2D materials engineering, photocatalysis, **synergic photocatalytic–biodegradation**, **biofilm engineering**, and advanced water treatment technologies. Demonstrated high-impact publications on self-powered optoelectronic devices and sustainable, synergic treatment of emerging contaminants in water, including work published in the prestigious ***Water Research*** journal.

Currently serving as Senior Technology Scientist at Inphlox Water Pvt. Ltd. (since Aug 2025), developing industrial-scale electrochemical wastewater treatment innovation. Rapidly developed deep reactor-technology expertise, introduced materials engineering and nanotechnology strategies, and drove concept-to-commercialization development through experimentation, validation, reactor innovation, and process optimization. Achieved **~80% OPEX reduction with ~90% improvement in pollutant removal capacity, while leading technical development and filing a patent within 7 months.**

With proven technical, interdisciplinary, collaborative, problem-solving, critical thinking and decision-making skills, ***I aim to innovate synergic, efficient and fit-for-purpose environmental remediation technology targeting the water-energy nexus, with a critical focus on sustainability.***

EXPERIENCE

Senior Technology Scientist

Inphlox Water Pvt. Ltd.

08/2025 Thane, India

The company develops and implements electrochemical water treatment technologies at commercial scale.

- Identified and optimized key operational parameters limiting the efficiency of electrochemical based water treatment processes.
- Integrated **materials engineering and nanotechnology** into electrochemical treatment systems, achieving ~80% OPEX reduction and ~90% higher pollutant removal
- **Led technical development, documentation and filed a patent within 7 months of joining.**

JSPS Postdoctoral fellow

School of Environment & Society, Tokyo Institute of Technology

06/2023 - 06/2025 Tokyo, Japan

Led research project for developing sustainable and synergic water treatment technology for emerging contaminant remediation

- Published fundamental mechanism governing **intimately coupled photocatalytic biodegradation (ICPB)** in the prestigious ***Water Research*** journal
- Achieved complete degradation of antibiotic pollutant with ***9 & 13 times faster kinetic rate constant*** than photocatalysis and biodegradation techniques.
- **Reduced electrical energy per order** by ***90%*** than photocatalysis technique

STRENGTHS

Technical expertise

Successfully developed individual and synergic water treatment techniques for enhanced degradation of emerging contaminants.

Interdisciplinary Research

Successfully integrated chemistry, materials engineering, graphene technology, photocatalysis and bio-electrochemistry towards developing sustainable water treatment technology.

Rapid Innovation & Intellectual Property Development

Led end-to-end technical development and documentation, rapidly translating lab innovations into intellectual property with a patent filed within 7 months of joining the industry.

SKILLS

- Chemistry lab techniques
- IP generation
- Water treatment technologies
- Interdisciplinary research
- Collaboration
- Project development
- Technical Presentations
- Communication
- Academic writing
- Leadership
- Teamwork
- Mentoring

RESEARCH GRANTS

JSPS International Postdoctoral grant (2023-2025)

School of Environment and Society, Institute of Science Tokyo (Formerly known as Tokyo Institute of Technology). Tokyo, Japan.

Project Title: Graphene-induced charge separation for enhancing intimately coupled photocatalytic biodegradation.
(Research grant – 2 million JPY)

IOE-IISc Postdoctoral fellowship

SusPaani Lab, Centre for Sustainable Technologies, Indian Institute of Science Bangalore, India.

Project Title: Multifunctional biocarrier-based intimately coupled photocatalytic biodegradation (ICPB) for enhanced degradation of organic micropollutants.
(Research grant – 2 Lacks INR)

EXPERIENCE

IOE-IISc Postdoctoral fellow

Centre for Sustainable Technologies, Indian Institute of Science, Bangalore.

04/2022 - 03/2023 Bengaluru, India

Led research project for developing sustainable technology for organic micropollutant remediation

- Developed ***graphene based composites*** for visible light induced photocatalysis degradation of antibiotic pollutant
- Developed ***working protocol*** for intimately coupled photocatalytic biodegradation water treatment technique
- Established inter and intra departmental collaboration

Technical Support and Application Specialist

Mettler Toledo Pvt. Ltd.

10/2021 - 03/2022 Mumbai, India

Technical support for electrochemical analytical instruments to Japan region

- Developed a 3D printed accessory of autosampler instrument for Japanese pharmaceutical client

Junior Research Fellow

Walchand Centre for Research in Nanotechnology and Bionanotechnology

02/2016 - 11/2017 Solapur, India

Nanotechnology Research Centre

- Developed green-adsorbents from agro-waste derived carbon materials for treatment of locally generated textile industry wastewater.
- Independently taught M.Sc. Nanotechnology course work including lectures and practicals.

R & D coordinator

Hindustan Unilever Ltd.

09/2015 - 01/2016 Mumbai, India

- Analyzed and developed reports on commercial air-purifier units.

EDUCATION

Doctor of Engineering, Department of Physical Sciences and Engineering

Nagoya Institute of Technology, Japan.

04/2018 - 03/2021 Nagoya, Japan

Master of Science in Analytical Chemistry

R. Ruia College, Mumbai University

06/2013 - 04/2015 Mumbai, India

Bachelor of Science in Chemistry

R. Ruia College, Mumbai University

06/2010 - 04/2013 Mumbai, India

WATER TREATMENT SKILLS

Water treatment:

Sustainable and energy-efficient treatment of emerging contaminants in water matrix using individual techniques i.e. **adsorption, photocatalytic degradation**, **biodegradation** and synergic techniques i.e. **intimately coupled photocatalytic biodegradation** (ICPB)

Bioreactor design

Designing bioreactor setup of ICPB technique for decentralized and continuous mode of operation.

Bio-electrochemical analysis

Photo-electrochemical and photo-bio-electrochemical characterization (**LSV, EIS, chrono-amperometry**) of photocatalysts and photocatalyst/biofilm composites

SELECTED PUBLICATION

Interface dependent electron shunting in graphene-integrated intimately coupled photocatalytic biodegradation

Water Research

Ajinkya Kishor Ranade, Akira Yamaguchi, Masahiro Miyauchi, Sreenivasan Ramaswami, Chihiro Yoshimura**

01/2025

https://doi.org/10.1016/j.watres.2024.123064

We reported a sustainable, energy-efficient strategy for degrading antibiotic pollutants, demonstrating that photoelectron transfer to the biofilm is a fundamental mechanism governing ICPB and can be controlled by the quality of the photocatalyst–biofilm interface.

Ultraviolet light induced electrical hysteresis effect in graphene-GaN heterojunction

Applied Physics Letters

Ajinkya K. Ranade, Rakesh D. Mahyavanshi, Pradeep Desai, Masashi Kato, Masaki Tanemura, Golap Kalita**

03/2019 https://doi.org/10.1063/1.5084190

The work was highlighted in renowned scientific websites, i.e., www.graphene-info.com and www.nanowerk.com, which report significant advancements in graphene technology and nanotechnology

1. https://www.graphene-info.com/understanding-graphenegan-and-other-2d3d-interfaces-uv-illumination-could-be-crucial-next-gen

2. https://www.nanowerk.com/nanotechnology-news2/newsid=52924.php

Formation of effective γ-CuI-GaN heterojunction with excellent ultraviolet photoresponsive photovoltage

Phys. Status Solidi A

Ajinkya K. Ranade, Pradeep Desai, Rakesh Mahyavanshi, Masaki Tanemura, Golap Kalita**

06/2019 **https://doi.org/10.1002/pssa.201900200**

MENTORSHIP

Throughout my career, I have been dedicated to mentoring and supervising students, fostering their development in water treatment technologies, materials engineering and carbon nanotechnology.

1. Sakura Science Program (Japanese student exchange program) 01/2021

Nagoya Institute of Technology – Indian Institute of Technology Kharagpur student exchang

2. Sakura Science Program (Japanese student exchange program) 01/2020

Nagoya Institute of Technology - Universiti Putra Malaysia student exchange

WATER TREATMENT SKILLS

Analytical techniques

Chemical analysis: Chemistry based qualitative and quantitative analysis techniques, COD, BOD, total organic carbon (TOC) and high-performance liquid chromatography (**HPLC**)

Material characterization: **UV-VIS-IR** spectroscopy with **DRS**, **FTIR** spectroscopy, **Raman** spectroscopy, X-ray diffraction (**XRD**), X-ray photoelectron spectroscopy (**XPS**), scanning electron microscope (**SEM-EDS**), atomic force microscopy (**AFM**), surface area measurement (**BET**), photoluminescence (**PL**) spectroscopy, semiconductor measurement (**I-V characterization**)

ADDITIONAL PUBLICATIONS

Characteristics of vertical Ga2O3 Schottky junctions with the interfacial hexagonal boron nitride film

ACS Omega

Rama, Venkata Krishna Rao, Ranade, Ajinkya, Desai, Pradeep, Todankar, Bhagyashri, Kalita, Golap, Suzuki, Hiroo, Tanemura, Masaki; Hayashi, Yasuhiko**

07/2022 https://doi.org/10.1021/acsomega.2c00506

Upcycling the barbeque grease into carbon nanomaterials

Carbon Trends

K.F. Chan, A.K. Ranade*, P. Desai*, M.A. Hazan*, N.F.I.Azman*, S. Shaharifin*, G. Kalita*, M.S. Mamat*, M. Tanemura*, Y.Yaakob**

12/2021 https://doi.org/10.1016/j.cartre.2021.100143

Influence on Electrochemical Reactivity and Synthesis of Stainless Steel/Nitrogen-Doped Carbon Nanofibers

J. Phys. Chem. C

Todankar, B., Desai, P., Ranade, A. K., Yaakob, Y., Asaka, T., Kalita, G., Tanemura

11/2021 https://doi.org/10.1021/acs.jpcc.1c06234

Temperature-dependent device properties of γ-CuI and β-Ga2O3 heterojunctions

SN Appl. Sci.

*Rama Venkata Krishna Rao, Ajinkya K. Ranade, Pradeep Desai, Golap Kalita, Hiroo Suzuki, Yasuhiko Hayashi**

09/2021 https://doi.org/10.1007/s42452-021-04774-3

Trifunctional Electrocatalytic Activities of Nitrogen-Doped Graphitic Carbon Nanofibers Synthesized by Chemical Vapor Deposition

Chemistry Select

Bhagyashri Todankar, Pradeep Desai, Ajinkya K. Ranade, Tharangattu N. Narayanan, Masaki Tanemura, Golap Kalita*,*

05/2021 https://doi.org/10.1002/slct.202101068

Synthesis of MoS2 Layers on GaN Using Ammonium Tetrathiomolybdate for Heterojunction Device Applications

Crystal Research and Technology

Pradeep Desai, Bhagyashri Todankar, Ajinkya K. Ranade, Masaharu Kondo, Takehisa Dewa, Masaki Tanemura, Golap Kalita**

01/2020 https://doi.org/10.1002/crat.202000198

Ultraviolet radiation-induced photovoltaic action in γ-CuI/β-Ga2O3 heterojunction

Materials Letters

Muhammed Emre Ayhan, Mandar Shinde, Bhagyashri Todankar, Pradeep Desai, Ajinkya K. Ranade, Masaki Tanemura, Golap Kalita*

11/2020 https://doi.org/10.1016/j.matlet.2019.127074

MENTORSHIP

3. Mr. Vyankatesh Kudkyal

2016 - 2017

Institution: Walchand College of Arts and Science, Solapur, India.

Topics: Wastewater purification using carbon-based smart filters. (He won 2nd prize in AAVISHKAR, a state-level student Research competition in Maharashtra, India, for a project conducted under my co-mentorship)

Current position: Ph.D. student at Chubu University, Japan

4. Ms. Su Rui

2024 - 2025

Institution: Institute of Science Tokyo, Japan.

Topics: Material fabrication and characterization, graphene interface engineering, photocatalysis and water treatment technologies (intimately coupled photocatalytic biodegradation)

Current position: MS student at Institute of Science Tokyo, Japan.

5. Ms. Farideh Malekdar,

2024 - 2025

Institution: Institute of Science Tokyo, Japan.

Topics: Material fabrication and characterization, Graphene technology, Photocatalysis.

Current position: Ph.D. student at Institute of Science Tokyo, Japan.

LANGUAGES

English
Proficient

Japanese
Intermediate

Marathi
Native

Hindi
Native

FIND ME ONLINE

Homepage
https://ajinkya-ranade.github.io/

LinkedIn
www.linkedin.com/in/ajinkya-ranade-phd-b4b899130

Google Scholar
https://scholar.google.com/citations?hl=en&user=Yir-Mq8AAAAJ

ADDITIONAL PUBLICATIONS

Growth of uniform MoS2 layers on free-standing GaN semiconductor for vertical heterojunction device application

Journal of Materials Science: Materials in Electronics

Pradeep Desai, Ajinkya K. Ranade, Mandar Shinde, Bhagyashri Todankar, Rakesh D. Mahyavanshi, Masaki Tanemura, Golap Kalita**

02/2020 https://doi.org/10.1007/s10854-019-02723-w

Influence of MoS2-silicon interface states on spectral photoresponse characteristics

Phys. Status Solidi A

Pradeep Desai, Ajinkya K. Ranade, Rakesh Mahyavanshi, Masaki Tanemura, Golap Kalita**

06/2019 https://doi.org/10.1002/pssa.201900349

Observing charge transfer interaction in CuI and MoS2 heterojunction for photoresponsive device application

ACS Appl. Electron.Mater.

Rakesh D. Mahyavanshi, Pradeep Desai, Ajinkya Ranade, Masaki Tanemura, Golap Kalita**

02/2019 https://doi.org/10.1021/acsaelm.8b00069

Photovoltaic action with broadband photoresponsivity in germanium-MoS2 ultrathin heterojunction

IEEE Transaction on electron devices

Rakesh D. Mahyavanshi, Golap Kalita*, Ajinkya Ranade, Pradeep Desai, Masaharu Kondo, Takehisa Dewa, Masaki Tanemura*

08/2018 10.1109/TED.2018.2864174

Photovoltaic action in graphene–Ga2O3 heterojunction with deep-ultraviolet irradiation

Phys. Status Solidi RRL

Golap Kalita, Rakesh D.Mahyavanshi, Pradeep Desai, Ajinkya K. Ranade, Masaharu Kondo, Takehisa Dewa, Masaki Tanemura*

05/2018 https://doi.org/10.1002/pssr.201800198

MOTIVATION OF RESEARCH

"Paani to Paani hai.....Paani Zindagani hai"
"Water is everything..... water is life itself.

-Snipet from a poem by Amol Palekar

REFERENCES

Prof. Chihiro Yoshimura

School of Environment and Society, Institute of Science Tokyo, Japan
(yoshimura.c.aa@m.titech.ac.jp ; yoshimura.c.8503@m.isct.ac.jp)

Dr. Sreenivasan Ramaswami

Centre for Sustainable Technology,
Indian Institute of Science, Bengaluru, India (sreeni@iisc.ac.in)

Dr. Golap Kalita

Principal Researcher,
Nippon Denko Co Ltd, Japan. (golapkalita@gmail.com)